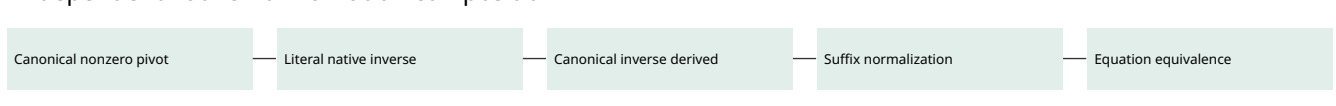
Independent native normalization composition review

Accepted the composed finite row contract at published revision 04e7fbe8d8. The literal narrow native inverse supplies its own canonical-unit and decoded-inverse facts; suffix normalization then yields canonical output, a zero prefix, pivot one, equivalent augmented equations and exact residual scaling.



The retained hypotheses are a prime modulus at most 2^32, a canonical row, a nonzero native pivot word and zeros before the pivot column. These are explicit inputs, not conclusions of this component. The inverse model preserves the low-bit test, 64-bit multiplication and unconditional final square already reviewed in the predecessor.

One independent wrapper passed in 24.469279 seconds under a 300-second total ceiling, 8 GiB address-space limit and one Lean worker. It freshly compiled both elimination precursors, Narrow, Normalization and Composition, then producer controls, named readbacks and reviewer controls. All 28,797 recursive dependency/toolchain hashes matched the producer pins and remained unchanged after verification.

Eight producer controls and four reviewer controls passed. The latter checked composite-modulus failure, zero-pivot failure, unit-row identity and retention of a noncanonical prefix word. Fifteen named declarations had only propext, Classical.choice and Quot.sound axioms, or no axioms. Three malformed audit controls were rejected. The deliberately false normalized-pivot-equals-two claim failed with Lean exit 1 as expected.

Source review matches native unit capture and the suffix loop including the right-hand-side column. Zero-prefix assumptions justify full-row scalar semantics. Pivot search, swaps, prefix propagation through subsequent iterations, array mutation, operation counters and the full solve4 return remain separate proof obligations.

Independence: reviewer did not author this source but knew the reviewed precursor contracts. The plan was frozen before reading this producer proof. This is manual native-source correspondence plus a pinned Lean rebuild, not an independent compiler or binary-kernel verification. Private elimination/graph/comparator snapshots and pinned shared caches were inventoried before and after; transient concurrent changes are not excluded by those inventories.